Bryan Widjaja

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Education

BINUS University — Alam Sutera, Tangerang
Undergraduate Computer Science - GPA : 3.97 — *Aug. 2024 - Current*

Experience

Part-Time Laboratory Assistant — Sep. 2025 - Current
BINUS University — *Alam Sutera, Tangerang*

- Instructed 100–120 students per semester across 4–5 laboratory sections, providing hands-on guidance and technical support
- Administered and proctored examinations, designed practical assessment cases, and graded student work
- Taught a broad range of programming and computer science topics, including Python, Computer Vision, C, C++, Data Structures with C, Java, Java Design Patterns, Enterprise Data Warehousing, and web development with HTML, CSS, and JavaScript

Projects

A Lightweight, Entropy-Aware Byte-Image CNN for Static Malware Detection — 2026
Python, PyTorch, Computer Vision — GitHub

- Developed a static malware classifier for Windows PE files using raw byte-level representations, requiring no execution of analyzed samples
- Designed a 64×64 byte-image pipeline combining byte values, local entropy, and PE section metadata with learned byte embeddings and Hilbert-curve sampling
- Implemented and evaluated a lightweight attention CNN with calibrated thresholds, achieving 0.988 ROC-AUC and 1.14% false-positive rate on family-disjoint evaluation
- Research paper accepted for presentation at the 2nd International Conference on Artificial Intelligence for Learning and Optimization (ICoAILO 2026)

Btardew Walley: A Terminal Farming Simulation Game — 2026
Java, OOP, Design Patterns, MVVM Architecture — GitHub

- Built a console-based farming and livestock simulation in Java with user authentication, tile-based maps, a day-cycle system, crop and animal lifecycles, a stores-and-inventory economy, and file-based save persistence
- Structured the codebase with an MVVM architecture split into model, view, viewmodel, service, and repository layers, organized as 111 classes across 32 packages
- Applied 8 GoF design patterns: Facade, Strategy, Command, Builder, Factory Method, Template Method, Iterator, and Composite

InstaTags: AI-Powered Instagram Hashtag Generator (Frontend + Chrome Extension) — 2026
SvelteKit, TypeScript, Tailwind CSS, Chrome Extensions — Frontend | Extension

- Developed an AI-powered web application for generating relevant Instagram hashtags from images
- Built a Manifest V3 Chrome extension that extracts images from active Instagram tabs using the Chrome Scripting API, sends them to the hashtag-generation API, and displays the generated tags for copying
- Developed reusable Svelte components and responsive pages for the landing page, image upload flow, usage guides, and product information using TypeScript, Tailwind CSS, and Siema

Technical Skills

Programming Languages: Python, Java, TypeScript, PHP
Frontend: React, Svelte
Backend: MySQL, PostgreSQL, Spring Boot
Machine Learning / Data: PyTorch, scikit-learn, OpenCV (cv2), pandas, NumPy, Matplotlib
Developer Tools: Git, Docker, Figma

Awards

Mentoring Scholarship — BINUS University
Taught undergraduate Calculus through BINUS University's SASC Tutoring Program, earning a one-semester scholarship — Feb. 2026